# MATTHEW POOL

## Software Engineer | Product Designer

*UI/UX | Native Mobile | Full-Stack Web | AI/ML | B2B/B2C*

mathyou.me@gmail.com 903-576-0223 Winnsboro, TX

matthew-pool.github.io

## PROFESSIONAL SUMMARY

Summa Cum Laude Computer Science graduate and Software Engineer with a rare background spanning end-to-end product development, 5+ years in Fortune 500 B2B/B2C sales, and a decade of technical support. I bridge the gap between UI/UX design and technical implementation—architecting backends, designing in Figma, and shipping to production—to deliver accessible, scalable software that drives real business value and solves real user problems.

## EDUCATION

### Bachelor of Science in Computer Science

*Southern New Hampshire University* | Sep 2021 – Dec 2024

🏆 **Summa Cum Laude** • ***3.99 GPA*** • **President's List**

📜 ***Certificate in Data Analysis*** • *uCertify (2024)*

### Associate of Arts in General Studies

*Tyler Junior College* | Aug 2003 – May 2005

# TECHNICAL SKILLS

### 🎨 Product Design

UI/UX Design, Figma, Inkscape, GIMP, Adobe Creative Suite, CAD, SketchUp, Responsive Design, User Flows, Information Architecture

### 💻 Programming Languages

Python, JavaScript, TypeScript, Kotlin, Java, C/C++, C#, SQL

### 📱 Mobile Development

Android SDK, Jetpack Compose, Material Design 3, MVVM Architecture, Coroutines, Hilt, Firebase, REST APIs

### 🌐 Web Technologies

React, Angular, Node.js, Express.js, RxJS, HTML5, CSS, SVG Animation, Vite, Bootstrap, Tailwind CSS, Spring Boot

### 🗄️ Databases

MySQL, MongoDB, Mongoose (ODM), AWS DynamoDB, Firebase Firestore

### 🤖 Data Science & Machine Learning

Keras (TensorFlow), scikit-learn, Pandas, NumPy, Matplotlib, DNN, CNN, Reinforcement Learning (RL), Deep Q-Networks (DQN)

### 🎮 Gaming & Graphics

Unity, OpenGL/GLSL, Game Development, Event-Driven Architecture, Game Loop Logic

### 🧪 Testing & QA

Postman, pytest, JUnit, MockK, Mockito, Espresso, RoboElectric, Jacoco, Jest, Selenium

### ⚙️ DevOps & Security

Docker/Docker Compose, AWS, CI/CD, Git/GitHub Actions, JWT, OWASP, Snyk, Gradle, Maven

## TECHNICAL PROJECTS

| | | |
|---|---|---|
| **Flick (Android)** | *Kotlin, Jetpack Compose, MVVM, Hilt, Firebase, GitHub Actions* | Proprietary ML recommendation engine, dual-axis gesture navigation, asynchronous preloading and caching for a perceived real-time streaming experience, and a CI/CD pipeline via GitHub Actions. |
| **Wormhole** | *Python, OOP, State Management* | Architected a complex interactive Python terminal game utilizing advanced object-oriented programming, dynamic state management, and robust game loop logic. |
| **Full-Stack Travel Platform** | *Angular, TypeScript, Docker, AWS DynamoDB* | Engineered an Angular SPA with a Dockerized backend, orchestrating a full DB migration from MongoDB to AWS. |
| **The Existence Paradox** | *React, TypeScript, Tailwind CSS* | Interactive web application featuring dynamic state management and a complex decision-tree navigation system. |
| **Financial Security Audit** | *Java, Spring Boot, OWASP, AES-256* | Refactored a legacy REST API to implement encryption at rest, mitigating injection vulnerabilities to achieve OWASP compliance. |
| **Machine Learning Models** | *Python, Keras (TensorFlow), scikit-learn, NumPy* | Trained and tuned a CNN for image classification and a Deep Q-Network (DQN) for reinforcement learning pathfinding. |
| **Product Design & UX** | *Figma, Wireframes, Prototyping* | End-to-end UX design for mobile and web apps: personas, wireframes, hi-fidelity interactive prototypes. |
| **Professional Portfolio** | *HTML5, CSS3, Vanilla JS, GitHub Actions* | Designed and hand-coded a zero-dependency responsive portfolio featuring a custom dark mode, complex SVG animations, and an automated CI/CD pipeline. |

*More projects and details at **matthew-pool.github.io***

## PROFESSIONAL EXPERIENCE

### Lead Software Engineer & Product Designer

*Jan 2025 – Present*

*AppBuddy (Independent Venture)*

- **Mobile Product Strategy:** Architected and launched **Flick**, a native Android video streaming platform—designed in Figma, built in Kotlin (MVVM), and deployed to Google Play with a proprietary ML recommendation engine.
- **Backend & Logic Engineering:** Architected **Wormhole**, a Python-based application, utilizing advanced object-oriented programming (OOP), state management, and complex logic design.
- **Web & Frontend Architecture:** Engineered a zero-dependency personal portfolio platform utilizing pure HTML/CSS/JS, and built **Existence Paradox**, a React/TypeScript application featuring dynamic state management and complex decision-tree routing.
- **DevOps & QA:** Built robust CI/CD pipelines via GitHub Actions automating unit (JUnit/MockK) and UI tests (Espresso) to ensure stable production deployments.

### Outbound Sales & Account Management (2004–2009)

⭐ Recurring Top Performer

**Sales Representative** • T.C.I.M. Services (B2B/B2C Outbound Sales)

- Conducted high-volume outbound sales calls and account management on behalf of Fortune 500 clients including American Express, AT&T, QVC, Citibank, Blue Cross Blue Shield, among others.
- Consistently ranked among the top performers in a rolling pool of 100+ representatives, exceeding KPIs across acquisition, upsell, and retention metrics.
- Developed call scripts adopted across the team and actively trained junior sales representatives on objection handling and closing techniques.

### Technical Support, IT Operations & Inbound Sales (2011–2013)

**Technical Support Specialist & Desktop Support** • Sutherland | New Electronics Warranty | TeleNetwork

- Diagnosed and resolved complex hardware, software, VoIP, and ISP issues via remote desktop tools; documented resolutions in CRM knowledge bases to improve team efficiency.
- Translated complex technical issues into clear guidance for diverse consumer audiences.
- Identified upsell and upgrade opportunities during support interactions, consistently meeting or exceeding inbound sales and satisfaction targets.
- Clients included Frontier Communications and DirecTV premium support customers.

### Shift Management & Customer Service (1997–2003 / 2009–2010) 👔 Management & Ops

**Shift Manager & Retail Associate** • Kmart / Fast Food & Customer Service (East TX)

- Directed workflow, delegated tasks, and trained new hires across fast-paced retail and food service environments.
- Regularly trusted to close stores independently at night, maintaining full financial accountability for cash registers and daily deposits.

### Independent Projects & Family Leave (2014–2021)

**Construction & Primary Caregiver**

- Designed, constructed, and converted a small woodshed intoa 3-bedroom residential structure - all from raw materials, independently managing resource allocation and electrical, plumbing, and finish work.
- Served as primary caregiver for a family of four while facilitating spouse's successful return to college.